

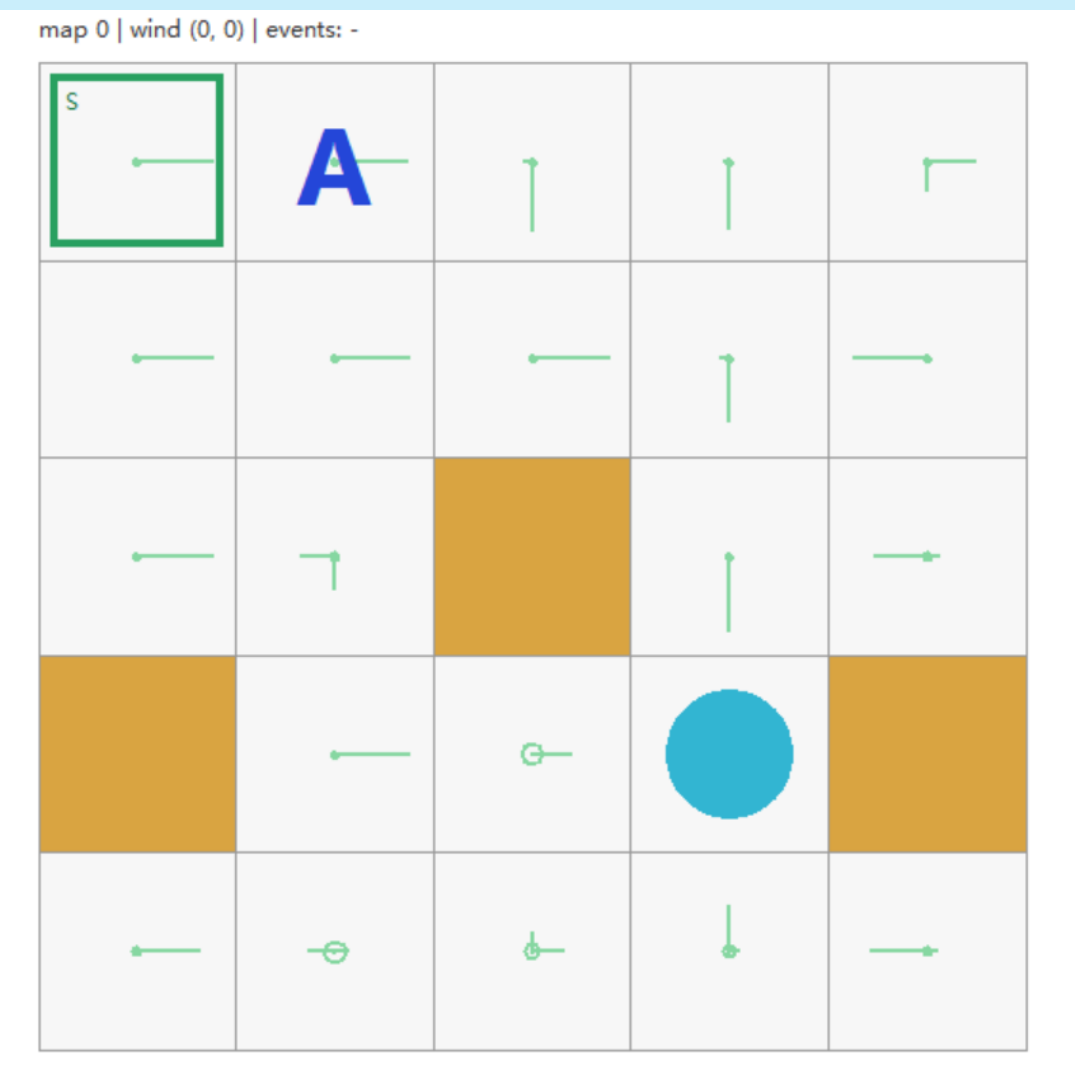
Feature-wise Step-Size Adaptation for Continual Reinforcement Learning in Non-Stationary Gridworlds

Team Members: Fandi Gou, Yidan Wu, Cunyang Xu, Ziheng Wang

Keywords: Continual RL; Feature-wise Step Sizes; Stochastic Meta-Descent



Individualized GridWorld



Non-Stationarity:

- Moving Goal
- Seasonal Wind + Reward
- Obstacle-Map Switches
- Wind-Only Dynamics

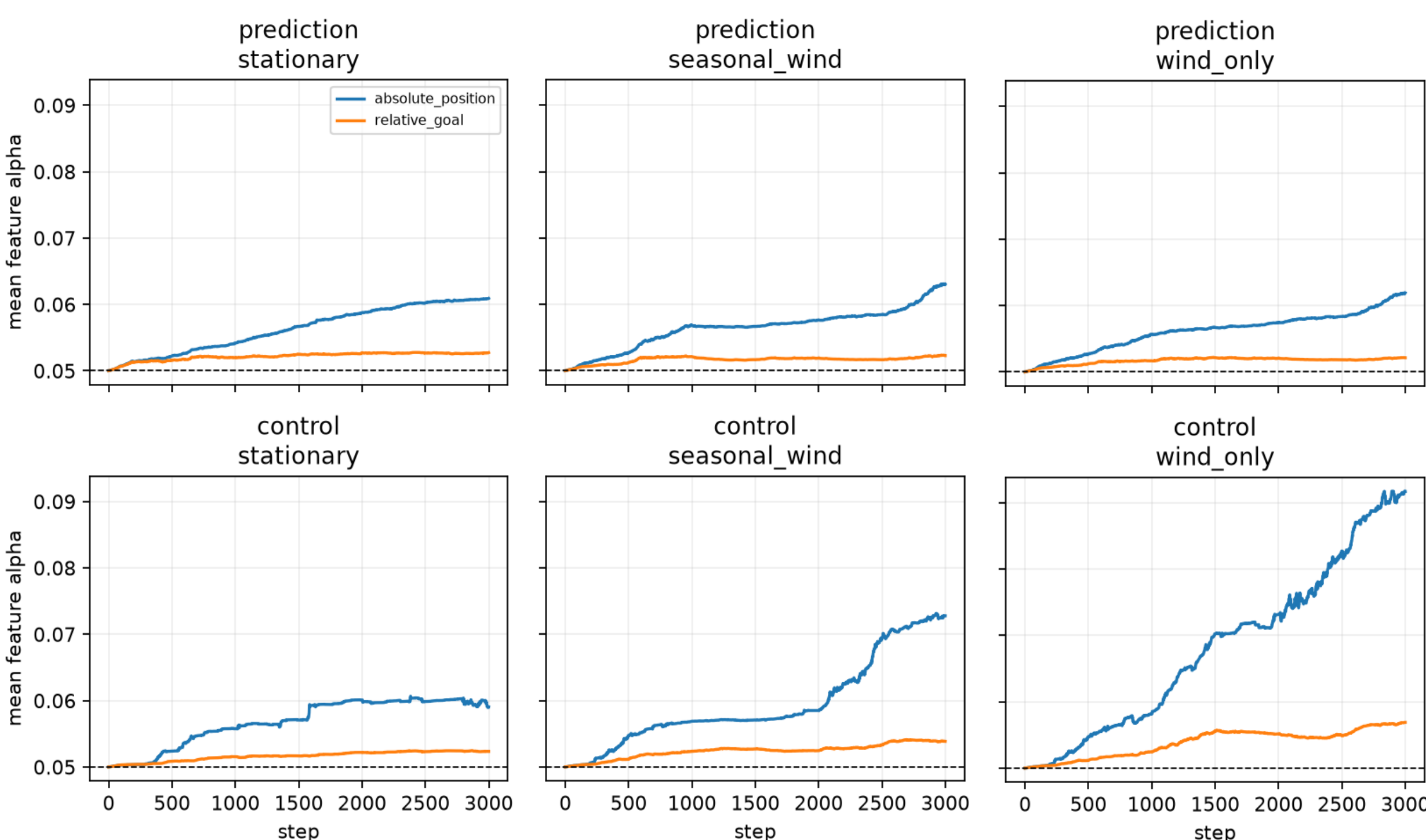
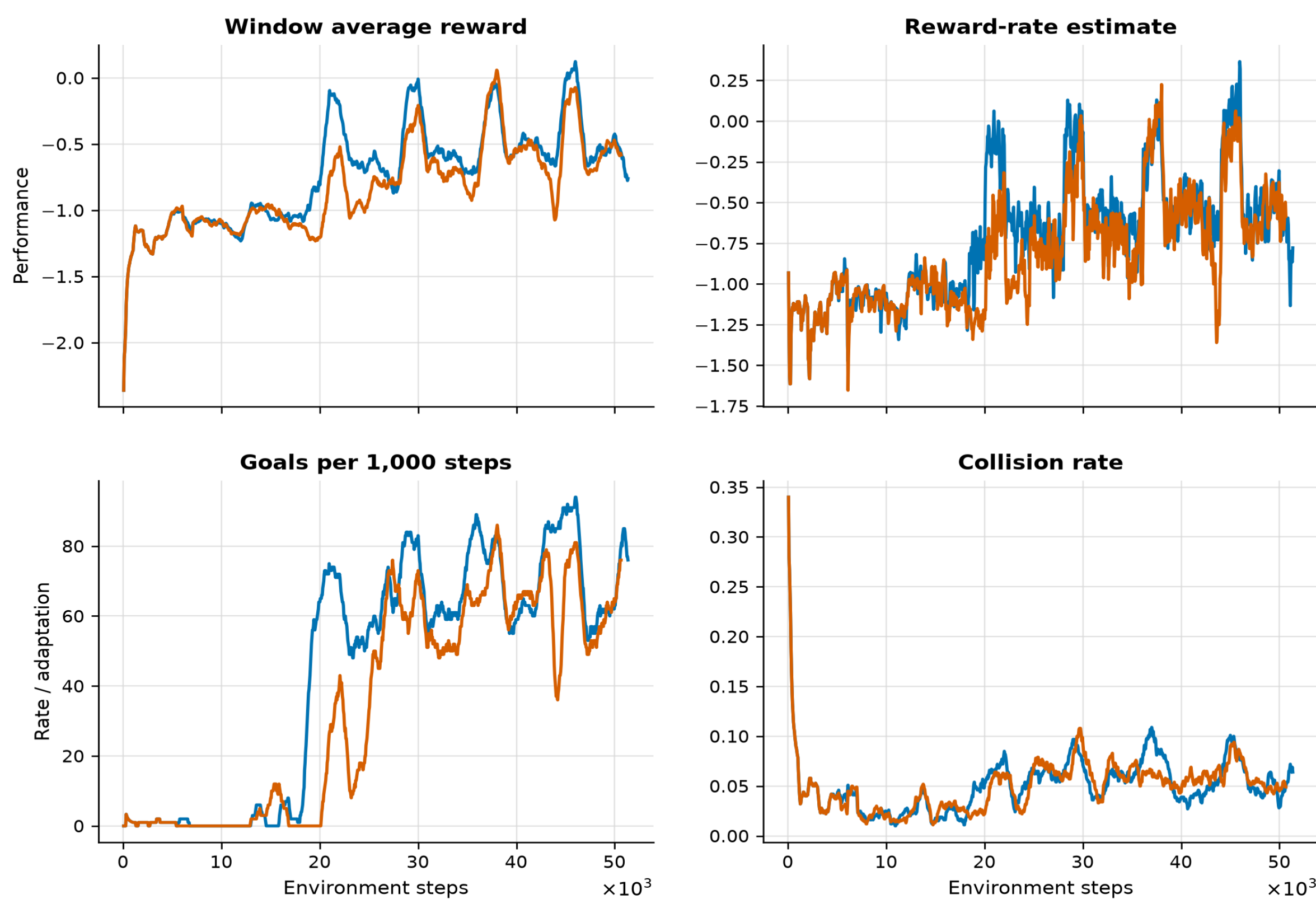
Setting:

- Random initial policy
- D = 55 Linear Features
- TD(0) / Sarsa(0)

Features (11/action): $\psi(s) = [1, \tilde{x}, \tilde{y}, \tilde{x}^2, \tilde{y}^2, \tilde{x}\tilde{y}, \tilde{d}_x, \tilde{d}_y, \tilde{d}_x^2, \tilde{d}_y^2, \tilde{d}_x\tilde{d}_y]$

Streaming RL metrics comparison

— TIDBD (adaptive) — Sarsa (fixed step size)



Motivation & Approach

Problem: A global step size forces all features to share the same stability.

Method: Differential TD(0)/SARSA(0) with TIDBD learns a positive step size for every feature.

Question: Does feature-wise adaptation reduce performance loss and recovery time after structured environmental changes?

Feature-wise TIDBD(0): stochastic meta-descent

$$\delta_t = R_{t+1} - \hat{\rho} + Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)$$

$$Q(S_t, A_t) = w_t^\top \phi(S_t, A_t)$$

$$\beta_{i,t+1} = \beta_{i,t} + \theta \delta_t x_{i,t} H_{i,t}$$

$$\alpha_{i,t+1} = \exp(\beta_{i,t+1})$$

$$w_{i,t+1} = w_{i,t} + \alpha_{i,t+1} \delta_t x_{i,t}$$

$$H_{i,t+1} = H_{i,t} [1 - \alpha_{i,t+1} x_{i,t}^2]^+ + \alpha_{i,t+1} \delta_t x_{i,t}$$

$$\hat{\rho}_{t+1} = \hat{\rho}_t + \eta_{\bar{r}} \delta_t$$

$$\lambda = 0 \implies z_{i,t} = x_{i,t}$$

$$[u]^+ = \max(u, 0)$$

$\hat{\rho}$: estimation of average reward
 $\frac{\partial \hat{\rho}}{\partial \beta}$ in average reward setting necessary?

TIDBD update follows Kearney et al. (2018); δ uses our continuing average-reward formulation.

Observed Control Performance

- On the displayed long stream, TIDBD-SARSA enters the goal-area and high-reward regime *earlier*.
- The visible reward shows faster adaptation; late-stage reward and collision rates are broadly comparable.
- The advantage is phase-dependent and should not be interpreted as uniform dominance in every environment.

Feature-wise Step-Size Dynamics

- Starting from $\alpha_0 = 0.05$, the learned step sizes separate by feature group across prediction and control.
- Absolute-position features increase more significantly than relative-goal features, especially under changing wind.
- This confirms active **feature-wise adaptation**; α alone is not a validated measure of feature importance.

Conclusions

- TIDBD provides a stable mechanism for increasing or decreasing each feature's step size online.
- It **accelerates adaptation** in the displayed non-stationary control stream, but aggregate rewards depend on the environment and metric.
- Step-size trajectories are useful diagnostics; feature selection still requires direct ablation and retention tests.

Reference: Kearney et al. (2018), TIDBD: Adapting Temporal-Difference Step-Sizes Through Stochastic Meta-Descent.

Contact information:

Fandi Gou (finley-gou3@sjtu.edu.cn); Yidan Wu (wyd510@sjtu.edu.cn);

Cunyang Xu (cunyang0925@gmail.com); Ziheng Wang (wzh0726@sjtu.edu.cn)